

RUSHI ASHOK SAVANI

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SUMMARY

AI/ML Engineer focused on building reliable AI systems across predictive modeling, intelligent retrieval, and production deployment. Experienced in developing end-to-end applications that combine machine learning, Large Language Model workflows, data pipelines, evaluation, and cloud infrastructure to solve real business problems. Strongest in turning messy, semi-structured operational data into scalable models, retrieval systems, and user-facing tools that improve decision-making, reliability, and business outcomes.

WORK EXPERIENCE

AI/ML Engineer Co-op

Jun 2025 – May 2026

ProHealth Group

Atlanta GA

- Orchestrated an agentic RAG workflow with LangChain on AWS Bedrock using Titan embeddings, OpenSearch, schema-aware retrieval, and Claude SQL generation to ground responses across business rules and unstructured notes in under 10 seconds.
- Engineered an XGBoost hospitalization-risk pipeline on clinical and claims data, combining patient-level temporal feature engineering, SMOTE-Tomek balancing, cross-validation, and threshold tuning to improve precision from 64% to 82%.
- Delivered a LLM-based anomaly investigation system on Vertex AI/Gemini for branch managers, using custom SQL rule pipelines and grounded clinical-data retrieval to identify operational-financial anomalies, surfacing ~\$70K in preventable loss.
- Improved LLM responses across RAG, SQL generation, and anomaly investigation by adding intent-routing logic, SQL/schema validation, citation checks, DeepEval regression tests, and prompt/caching optimizations to improve reliability.
- Productionized ML and LLM applications as Dockerized FastAPI services on AWS EC2, using S3, IAM, and GitHub Actions CI/CD to automate secure dev-to-production releases.

Data Science - Research Assistant

Jan 2025 – May 2025

Syracuse University

Syracuse, NY

- Built an automated NLP research pipeline for 5,000+ AI-adoption survey responses, using Python, pandas, SciPy, and Power BI to clean raw survey data, generate EDA summaries, and surface department-level usage, sentiment, and training-need patterns.
- Fine-tuned a RoBERTa-based sentiment classification model using Hugging Face Transformers and PyTorch, tracking experiments with MLflow and applying stratified validation to achieve 91% classification accuracy.
- Applied transformer-based embeddings and semantic clustering to analyze recurring themes to understand adoption challenges, employee concerns, and support requirements, providing findings that informed internal planning and decision-making.

Data Analyst

Jun 2023 - Jun 2024

Krishna Jewels

Mumbai, IN

- Designed and implemented the company's first analytics-ready data platform by modeling inventory, sales, supplier, and accounting data into a MySQL relational schema; migrated 40K+ historical records and improved data retrieval speed by 70%.
- Engineered automated SQL-based ETL pipelines and Power BI dashboards for sales, inventory, and margin analytics, transforming paper-based records into structured datasets with daily refreshes and eliminating ~15 hours/week of manual work.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, TypeScript, FastAPI, REST APIs, Git

Machine Learning: PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers, MLflow, Fine-tuning

GenAI & LLM Systems: LangChain, LangGraph, Prompt Engineering, Vector Databases, AWS Bedrock, Vertex AI, Anthropic Claude

Cloud & MLOps: AWS (EC2, ECS, S3, SageMaker), Microsoft Azure, Docker, GitHub Actions CI/CD

Data Engineering & BI: PostgreSQL, MySQL, Data modeling, Snowflake, DBT, Fivetran, ETL Pipelines, Power BI, Tableau

PROJECTS

PrivateRAG | *Private On-premises Q&A System*, [GitHub](#)

- Deployed a privacy-first RAG system for sensitive clinical and policy documents using FastAPI, LlamaIndex, local Llama 3.1 via llama.cpp/Ollama, HuggingFace nomic embeddings, and Qdrant vector search.
- Implemented document ingestion, vector indexing, contextual chat, streaming OpenAI-style API responses, source attribution, file-level retrieval filters, and Docker-based local deployment to keep data inside a zero-egress workflow.

KarmaScore Predictor | *Subscription Churn Analysis*

- Predicted subscriber churn across 30K+ records using XGBoost, narrowing 60 raw variables to 23 features through target encoding, recursive feature elimination, SHAP-based score, and multicollinearity reduction, achieving 89% precision.
- Mapped SHAP scores into a customer-level intervention likelihood metric ranking which at-risk subscribers are worth retaining based on projected profitability, displayed in an interactive Streamlit dashboard.

JobFlow AI | *LLM-Powered Assistant*, [GitHub](#)

- Launched a Gemini-powered application assistant with FastAPI, SQLAlchemy, and provider-based LLM orchestration to generate tailored resumes and cover letters, reducing manual application customization time from hours to minutes.
- Integrated production-style backend workflows for pipeline execution, user-scoped retrieval, resume upload validation, metadata persistence, SHA-256 file hashing, and Dockerized API deployment.

EDUCATION

Master of Science | Information Systems, Data Science

Syracuse University, School of Information Studies, New York

May 2026

(GPA: 3.9/4)

Bachelor of Engineering | Information Technology

Thadomal Shahani Engineering College, India

May 2024

(GPA: 8.52/10)